# WAHYU TRI WICAKSONO

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Electrical, Instrumentation and Control Engineer with hands-on experience in oil & gas and power plant operations, commissioning support, and industrial control systems. Experienced in plant shutdown and outage execution, electrical power system testing (LV/MV), generator and transformer protection relay configuration, and DCS/PLC troubleshooting. Strong background in power plant electrical systems, instrumentation, and process automation, with exposure to offshore oil and gas operations and DeltaV-based control systems. Highly motivated in commissioning and offshore energy projects within global industrial environments.

## PROFESSIONAL EXPERIENCE

**Pertamina Geothermal Energy, Lumut Balai – Onshore** *Dec 2025 – Present*

*Electrical, Instrument & Control Engineer (Maintenance Support and Project Team)*

- Supported commissioning, plant outage, and first year inspection activities including punch list clearance, NCR resolution, system performance verification, and technical coordination with EPC project teams.
- Provided engineering and technical support for Electrical, Instrumentation, and Control (EIC) systems in a 2×55 MW geothermal power plant, supporting plant operation, system modification, and project activities to ensure reliable and safe operation.
- Performed troubleshooting, modification, and performance improvement of DCS (TOSMAP, Ovation) and PLC systems (Siemens & Schneider), including logic modification, I/O expansion, HMI configuration, and system integration.
- Conducted feasibility studies and engineering analysis for control system integration and power interconnection between Unit-1 and Unit-2 to improve operational flexibility and system reliability.
- Performed testing, configuration, and maintenance of power system equipment and protection systems including MV/LV systems, transformers (6.3 kV, 13.8 kV, 150 kV GSUT), protection relays (SEPAM, Beckwith), UPS, and GCB equipment.
- Supported turbine control, AVR, Electro-Hydraulic Control (EHC), Balance of Plant (BOP), and package control systems including instrumentation calibration, valve testing, and process performance verification.

**ENI Indonesia, Balikpapan – Onshore & Offshore** *Jul 2025 – Dec 2025*

*Instrumentation & Automation Specialist*

- Provided support to the maintenance team for multiple offshore production assets, including the ORF and Jangkrik Floating Production Unit (FPU) and East Sepinggan, Muara Bakau, and West Ganal blocks, covering instrumentation and automation.
- Supported routine maintenance activities on metering systems (Ultrasonic Flow Meters, Coriolis Meters, and Multiphase Flow Meters), automation systems (ICSS – Schneider Quantum and Honeywell Experion 300), and DeltaV Metering Systems.
- Supervised vendor execution during maintenance, upgrades, turnaround, and overhaul activities, including review of Scope of Work, work breakdown, dismantling, Factory Acceptance Testing (FAT), Site Acceptance Testing (SAT), commissioning, startup, and final acceptance from an instrumentation and automation perspective.
- Planned and scheduled Preventive Maintenance (PM) and Corrective Maintenance (CM), while preparing inputs for the Work Program and Budget (WP&B) submissions to SKK Migas (Indonesian oil and gas regulator).
- Coordinated and planned recertification works for Ultrasonic Flow Metering Systems, as well as offshore critical safety systems such as High Integrity Pressure Protection System (HIPPS) and Pipeline Leak Detection System (PLDS).
- Reviewed technical feasibility and compliance of instrumentation and automation systems to ensure safety, reliability, and operational efficiency during offshore maintenance campaigns.

**PT. Control Systems Arena Para Nusa, Jakarta – Onshore & Offshore** *Sept 2024 – Jul 2025*

*Control System Engineer (DCS DeltaV Engineer)*

- Developed design documents, graphics, and control modules specifically for DCS DeltaV systems, ensuring seamless integration and optimal performance in various industrial processes.
- Executed Factory Acceptance Testing (FAT), Site Acceptance Testing (SAT), and site commissioning activities for Emerson DCS DeltaV and PLC RX3i systems, ensuring they meet the project's technical and operational requirements.
- Performed routine maintenance services on installed systems, ensuring reliability and minimizing downtime in user and operational environments.
- Provided on-call support for troubleshooting and resolving issues related to DCS DeltaV and PLC RX3i systems, maintaining high standards of system availability and performance.

**PT. Mandau Cipta Tenaga Nusantara, NDC Plant Pertamina Hulu Rokan, Riau – Onshore** *May 2024 – Aug 2024*

*Instrument and Control Engineer*

- Managed phase 3 upgrade and redundancy of the control system at the NDC Gas Turbine Plant, including Combustion Turbine Unit-2, Heat Recovery Steam Generator-2, and Balance of Plant DPU-22.
- Supervised the Network Team to improve communication within the Distributed Control System at NDC Plant, specifically for 2 DPU HRSG units, 4 DPU Electrical Package units, and Power Control Room Area.
- Participated in the Hot Gas Path Inspection team for Turbine Unit-2 and the Fire Alarm System inspection team at the NDC Plant.
- Supervised the improvement of the Fire Alarm System to integration with DCS Network Back Bone, especially for Electrical Package, Mechanical Package, Main Trafo and SSTX Trafo, Combustion Turbines and Generators.

**Freelance Engineer, Yogyakarta – Remote & Onsite** *Feb 2024 – Jun 2024*
*Hardware and Automation Engineer*
- Designed a control system for a bullet press machine using an Omron and Mitsubishi PLC with high accuracy.
- Developed an automation system for a press machine using pneumatics with a solenoid operating at pressures exceeding 10 bar.
- Designed and developed monitoring systems to measure vibrations in instrument equipment with an accuracy of up to 1 $m/s^2$.
- Integrated sensor systems with field devices and monitor displays, achieving a data transmission time of approximately 10 s.

**Faculty Engineering Gadjah Mada University, Sleman – Onsite** *Jan 2024 – Feb 2024*
*Hardware Software Specialist*
- Designed and developed an IoT-based monitoring system with a weighing capacity of up to 100 kg and 0.5% sensitivity.
- Integrated the waste weighing system with the database system.
- Designed database network routes and touch screen GUI for the waste weighing device.

**PT. Adil Makmur Sentosa, Jepara – Onsite** *Jun 2023 – Aug 2023*
*Electrical and Instrumentation Engineer Intern*
- Evaluated and analyzed electrical and instrumentation systems for processing domestic and medical waste using incinerators.
- Created technical documentation and conducted calculations for the operation of electric motor-based milling machines.
- Simulated and optimized the combustion process in the incinerator using Ansys software.

**PT. Sorini Agro Asia Corporindo Cargill, Pandaan – Onsite** *Dec 2022 – Mar 2023*
*Electrical, Instrumentation and Automation Engineer Intern*
- Observed environmental physical conditions of failed instruments to identify relationships between environmental conditions and instrument failure using descriptive analysis methods.
- Verified, evaluated, and revised calibration documents, as well as maintained instrument equipment.
- Proactively resolved issues during instrument maintenance, instrument drawing, and focused on automation and instrumentation.

## EDUCATION AND QUALIFICATIONS

**Bachelor Engineer in Engineering Physics, Gadjah Mada University** *Sep 2020 – Jan 2024*
*Cumulative GPA: 3.46/4.00 (147 credits)*
- Thesis titled "Hydrogen Production Process Monitoring System in Photovoltaic Electrochemical (PVEC) Reactor Based on IoT"

**Certificate in Instrument Inspector Level 1, instrument-inspector.com** *Jan 2026*
- Training for ITP, FAT, SAT, Instrument Standard and Test, Verification Method, Inspection Method and Acceptance Criteria.

**Certificate in Hazardous Area Classification – Electrical Equipment Selection, Engineering Academy** *Dec 2026*
- Fundamental of Explosion, Hazardous Area, Material Classification, and Electrical Equipment in Oil & Gas and Petrochemical.

**Certificate in BOSIET with CA-EBS Digital Delivery & TSbB, OPITO** *Aug 2025*
- Basic Offshore Safety Induction and Emergency Training with Compressed Air Emergency Breathing System & TSbB

**Certificate in Basic for Building Management Systems, Urcoursez Academy** *Apr 2024*
- Fundamental Building Management System (Basic Mechanical, Electrical and Plumbing).

**Certificate in Fundamental Piping and Instrumentation Diagram (P&ID), Engineering Academy** *Mar 2024*
- PFD, P&ID, 3D Overview of Industrial Plant, Piping Specification & Classification, Piping to Instrument Connection.

**Certificate in Field Instrument Engineer, Insys Exclusyve** *Feb 2023 – Mar 2023*
- Field Instrumentation Design – Foundation with Natural Gas Scrubber System Case Study.

**Certificate in Artificial Intelligence Mastery Program MSIB, PT Orbit Ventura Indonesia** *Aug 2022 – Dec 2022*
- Fundamental Artificial Intelligence, Data Analysis and Processing Capabilities Based on Machine Learning and Deep Learning.

## SKILLS & OTHERS

**Soft Skills** Leadership, Adaptability, Communication, Presentation, Teamwork, Time management, Attention to detail, Analytical Thinking, Problem-solving.

**Hard Skills** Instrumentation, Electrical, Programmable Logic Controller (PLC), Distibuted Control System (DCS), Maintenance, Network Communication System, Integrated Fire Alarm System (FAS), Operation and Control, Measurement System, Callibration, Software and Hardware, AutoCAD, Data Analyst, Microsoft Office, Pneumatics and Hydraulic System.

**Languages** Indonesian (Native), English (Professional), French (Basic).

**Achievements** The first winner of the Siaga Awards with Research about Hydrogen Technology, Bronze Medal class PKM- Riset Eksakta at PIMNAS (Pekan Ilmiah Mahasiswa Nasional) 35 2022, The first winner of the LKTI Agrocompetition 2022 competition with TABUR-21, Silver Medal for the project TABUR-21 at International Research Innovation for Agritech Development AGREETION 2022.

## PROJECT & DEVELOPMENT

**First Year Inspection Unit-2 55 MW Lumut Balai Geothermal Power Plant** *Feb 2026 – Present*
*PT. Pertamina Geothermal Energy Lumut Balai*
Supported the first major shutdown (First Year Inspection) of Unit-2 (55 MW), performing system verification, functional testing, and performance assessment of electrical, instrumentation, and control systems to validate equipment integrity and plant reliability.

**Plant Outage Unit-1 55 MW Lumut Balai Geothermal Power Plant** *Jan 2026*
*PT. Pertamina Geothermal Energy Lumut Balai*
Supervised and Lead team for planned plant outage activities for Unit-1 (55 MW), performing inspection, maintenance, and functional verification of electrical, instrumentation, and control systems to ensure reliable and safe plant operation.

**Replacement USM Flow Meter for Second Tie-In Onshore Receiving Facilities** *Sept 2025*
*Eni Muara Bakau B.V.*
Supervised the replacement and wet & dry calibration of Ultrasonic Flow Meter (USM) for the second tie-in system, ensuring metering accuracy, system integrity, and compliance with measurement and operational standards during project execution.

**ICSS Pulau Gading Booster Compressor Facilities** *Apr 2025 – Jul 2025*
*PT. Pertamina Hulu Energi Jambi Merang*
Dedicated Control Systems Engineer for installation of Integrated Control and Safety System DCS DeltaV and Tie in process with Honeywell Control System for Booster Compressor Unit in area PT. PHE Jambi Merang on the construction project (EPC) with PT. Indoturbine as Contractor Project.

**Base Oil Receiving Facility Revamp – RU IV Production Unit** *Mar 2025 – Jun 2025*
*PT. Pertamina Lubricants Cilacap*
Contributed control system upgrade and modernization of the base oil receiving facility using DeltaV DCS version 15, supporting system configuration, integration, and functional verification during facility revamp activities.

**ABG Development Project Existing Plant Modification – Join Engineering with Pamtek Engineering** *Mar 2025 – Jun 2025*
*PT. Pertamina Explporation and Production Regional 2, Zone 7, Asset 3 Akasia Bagus Existing Plant*
Led control system re-engineering and integration of existing PLC M580 BPCS and Safety SIL systems with the newly developed plant, ensuring seamless control integration and operational reliability during EPC plant modification.

**SCADA and Safety System Implementation – Pengapon–Boyolali Pipeline Project** *Dec 2024 – May 2025*
*PT. Pertamina Patra Niaga and PT. PGN Solution*
Contributed to implementation of SCADA and Safety Instrumented System (SIS) using DeltaV and SCADA platforms for the Pengapon–Boyolali pipeline, supporting system integration, functional testing, commissioning and operational safety.

**DeltaV Expansion Project for Bravo Platform Offshore North West Java** *Oct 2024 – Jul 2025*
*PT. Pertamina Hulu Energi ONWJ*
Designed and developed a process control system for the DeltaV DCS expansion project, integrating new control systems with the existing DCS across all areas of the Bravo Offshore Platform (ONWJ). The project involved full integration with package systems for oil and gas separation from three-phase fluids, covering process control systems for oil and gas treatment across multiple offshore production areas. Responsibilities also included engineering of process interlock systems and annunciator alarm systems to ensure safe and reliable operations.

**ICSS SP Akasia Bagus & Well Monitoring System – Tie in Existing ABG Development Plant** *Aug 2024 – Jul 2025*
*PT. Pertamina Exploration and Production Regional 2, Zone 7, Asset 3 Akasia Bagus & Tugu Barat*
Dedicated Instrument and Control Systems Engineer for installation of Integrated Control and Safety System DCS DeltaV and oil well monitoring system in PT. Pertamina Exploration and Production zone 7 area's on the construction project (EPC) with Tripatra Engineering Construction and Procurement as Main Contractor for Stasiun Pengumpul ABG Stage 1 di lapangan Akasia Bagus.

**Network Improvement, Unit-2 DCS Upgrade, CEMS HRSG, and Fire Alarm System** *May 2024 – Aug 2024*
*PT. PLN Mandau Cipta Tenaga Nusantara, NDC Plant – CGS 11 Pertamina Hulu Rokan*
Led the upgrade of Unit-2 DCS from Ovation to Honeywell, enhancing system reliability and performance. Supervised network communication improvements across 2 HRSG units and 4 electrical packages. Managed the upgrade of the Fire Alarm System, integrated it with the plant's central monitoring system, and improved the CEMS for HRSG for accurate emissions reporting. Conducted comprehensive testing to ensure compliance with industry standards.

**Automatic Water Pumping Unit and Level Control** *Mar 2024 – Des 2024*
*Vega Instrument and Engineering Physics, Universitas Gadjah Mada*
Designed and developed an automatic water pumping unit with level control for efficient water management. Implemented sensor-based monitoring to maintain optimal water levels, ensuring reliable operation and preventing overflow. Integrated system controls to automate water pumping based on real-time data, enhancing system efficiency and reducing manual intervention.

**Precision Machine Automation Engineering with PLC and Pneumatic Systems** *Feb 2024 – Jun 2024*
*Freelance Engineer Project, Sleman District Police*
Specialized in designing press machine control systems with Omron and Mitsubishi PLCs. Skilled in pneumatic systems, including simulations and implementations. Developed high-pressure automation systems using double-acting cylinders and solenoid valves.